

Institute of Actuaries of India

Subject CT3 – Probability & Mathematical Statistics

March 2018 Examinations

Indicative Solutions

Introduction:

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other approaches leading to a valid answer and examiners have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) k^{th} moment = $E[X^k]$ [0.5]
- ii) k^{th} moment about α . = $E[(X - \alpha)^k]$ [0.5]
- iii) k^{th} central moment = $E[(X - \mu)^k]$ where μ is the mean [0.5]
- iv) Coefficient of skewness = $\frac{E[(X - \mu)^3]}{\sigma^3}$ where σ is the std. deviation [0.5]
- [2 marks]

Solution 2:

	Count	Mean	Standard Deviation
Year 1	190	53	19
Year 2	175	57	15

$$\bar{X} = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2}{n_1 + n_2} = \frac{190 \times 53 + 175 \times 57}{190 + 175} = 54.9$$

$$\sum X_{1i}^2 = s.d._1^2 (n_1 - 1) + n_1 \bar{X}_1^2 = 19^2(190 - 1) + (190)53^2 = 601,937$$

$$\sum X_{2i}^2 = s.d._2^2 (n_2 - 1) + n_2 \bar{X}_2^2 = 15^2(175 - 1) + (175)57^2 = 607,725$$

$$s.d.^2 = (\sum X_{1i}^2 + \sum X_{2i}^2 - (n_1 + n_2) \bar{X}^2) / (n_1 + n_2 - 1) = 17.29^2$$

Hence, the std. deviation of the sample = 17.29

[5 marks]

Solution 3:

- i) $X \sim N(15, 5^2)$ and $Y \sim N(12, 10^2)$

Determine the distribution of $Z = X - Y + 5$

Mean of Z = Mean of X + Mean of Y + 5 = 15 - 12 + 5 = 8

Var[Z] = Var[X] + Var[Y] = 25 + 100 = 125

$Z \sim N(8, 11.18^2)$

[2]

- ii) The random variable Z follows $\sim N\left(8, \frac{11.18^2}{5}\right)$

$$\text{Hence } P(\bar{Z} > 10) = P\left(\frac{\bar{Z} - 8}{5} > 0.4\right) = 1 - \Phi(0.4) = 1 - 0.6554 = 0.3446$$

$$\text{Also } \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2. \text{ Hence } P(S^2 > 20^2) = P\left(\frac{(n-1)S^2}{\sigma^2} > \frac{4 \times 400}{125}\right)$$

$$\text{i.e. } P\left(\frac{(n-1)S^2}{\sigma^2} > 12.8\right) = P(\chi_4^2 > 12.8) = .0124$$

Since the mean and standard deviation of a normal population are independent, the probability that mean greater than 10 and standard deviation greater than 20 is

$$(0.3446)(0.0124) = 0.0043$$

[5]

[7 marks]

Solution 4:

$$\begin{aligned} \text{i) } P[\text{Withdrawal}] &= P[\text{Withdrawal} | \text{Agency}] \cdot P[\text{Agency}] + \\ &P[\text{Withdrawal} | \text{Bank}] \cdot P[\text{Bank}] + P[\text{Surrender} | \text{Online}] \cdot P[\text{Online}] \\ &= 0.05 \left(\frac{2000}{10000} \right) + 0.08 \left(\frac{3500}{10000} \right) + 0.14 \left(\frac{4500}{10000} \right) = 0.101 \end{aligned}$$

[2]

$$\begin{aligned} \text{ii) } P[\text{Withdrawal} \cap \text{Agency}] &= P[\text{Withdrawal} | \text{Agency}] \cdot P[\text{Agency}] = 0.05 * 0.20 = 0.01 \\ P[\text{Withdrawal} \cap \text{Bank}] &= P[\text{Withdrawal} | \text{Bank}] \cdot P[\text{Bank}] = 0.08 * 0.35 = 0.028 \\ P[\text{Withdrawal} \cap \text{Not online}] &= 0.01 + 0.028 = 0.038 \\ P[\text{Withdrawal} | \text{Not Online}] &= P[\text{Withdrawal} \cap \text{Not Online}] / P[\text{Not Online}] \\ &= \frac{0.038}{1 - 0.45} = 0.0691 \end{aligned}$$

[2]

$$\text{iii) } P[\text{Bank} | \text{Withdrawal}] = \frac{P[\text{Withdrawal} \cap \text{Bank}]}{P[\text{Withdrawal}]} = \frac{0.280}{0.101} = 0.2772$$

[2]
[6 marks]

Solution 5:

$$\begin{aligned} \text{i) Given that } X \text{ and } Y \text{ are iid from exponential distribution with mean } \frac{1}{2} \text{ i.e. } \sim \text{Exp} \\ F_Z(t) = P(Z \leq t) &= 1 - P(\text{Min}(X, Y) > t) \\ &= 1 - P(X > t \text{ and } Y > t) = 1 - P(X > t) * P(Y > y), \text{ since } X \text{ and } Y \text{ are independent} \\ &= 1 - \{1 - P(X \leq t)\} * \{1 - P(Y \leq t)\} = 1 - \{(1 - F_X(t)) * (1 - F_Y(t))\} \\ &= 1 - \{1 - (1 - e^{-2t})\} * \{1 - (1 - e^{-2t})\} \quad [\text{as } F_X(t) = F_Y(t) = 1 - e^{-\lambda t}] \\ &= 1 - e^{-4t} \end{aligned}$$

[3]

$$\text{ii) } 1 - e^{-4t} \text{ is the CDF of an exponential distribution; hence the mean is: } \frac{1}{4}.$$

[1]
[4 marks]

Solution 6:

- i) Let P_i : denote the premium for selling the i th insurance policy
 C_i : denote the commission for selling the i th insurance policy
 T : The total commission earned by an agent:

$$T = C_1 + C_2 + \dots + C_N = 0.3(P_1 + P_2 + \dots + P_N)$$

$$\text{Since } P_i \sim \text{lognormal}(6, 2.5) \quad E[P] = e^{(6+6.25/2)} = 9182$$

$$\text{var}(P) = e^{2 \times (6+6.25/2)} (e^{6.25} - 1) = (208779)^2$$

$$N \sim \text{Poi}(20) \text{ so we have } E[N] = 20 \text{ and } V[N] = 20.$$

$$\text{Hence } E[T] = E[C] \cdot E[N] = 0.3 \times E[P] \cdot E[N]$$

$$E[T] = 0.3 \times 9182 \times 20 = 55092$$

$$\text{var}[T] = E[N] \cdot \text{var}[C] + \text{var}[N] \cdot E[C]^2$$

$$= E[N] \times 0.09 \times \text{var}[P] + \text{var}[N] \times 0.09 \times E[P]^2$$

$$\text{var}[T] = 0.09(20 \times 208779^2 + 20 \times 9182^2) = 280378^2$$

[5]

ii) The assumptions are:

The commissions paid to agents are independent; and the sample size is very large.

The central limit theorem is applicable and that the overall outgo on commissions is normally distributed.

The mean of overall outgo is then $1000 \times 55092 = 55092000$

The variance of overall outgo is $1000 \times 280378^2 = 8866325^2$

Probability that total outgo exceeds INR 6.5 million is

$$1 - P(\text{total outgo} < 65000000) = 1 - \Phi\left(\frac{65000000 - 55092000}{8866325}\right)$$

$$= 1 - \Phi(1.1175) = 0.1318$$

[4]

iii) Let the proportion of the existing commissions to be paid is 'x'.

The revised mean of the normal distribution is thus $55092000x$

The revised variance of normal distribution is $(8866325x)^2$

At 95% level of significance, the value of std. normal variate is 1.6449

We need to find 'x' for which $\frac{65000000 - 55092000x}{8866325x} = 1.6449$

$x = 0.9329$ i.e., the revised commission is 93.29% of old commission.

Hence, the revised commission rate is $30\% \times 93.29\% = 28\%$

[5]

[14 marks]

Solution 7:

When $X_i \sim \text{Exp}(\theta)$, we have $M_x(t) = (1 - t/\theta)^{-1}$

For $Y = \sum_{i=1}^n X_i$, X_i are independent, we have

$$M_Y(t) = E[e^{tY}] = E[e^{t \sum_{i=1}^n X_i}] = \prod E[e^{tX_i}] = E[e^{tX}]^n$$

$$\text{giving } M_Y(t) = (1 - t/\theta)^{-n}$$

$$\text{Thus, } M_z(t) = \sqrt{M_Y(t)} = (1 - t/\theta)^{-n/2}.$$

The MGF of chi-squared distribution is $(1 - 2t)^{-n/2}$. Hence $M_z(t)$ is the MGF of a chi-squared distribution with 'n' degrees of freedom when $\theta = 0.5$.

[4 marks]

Solution 8:

i) The likelihood function for the given sample of observations is:

$$L(\alpha) = \left(\frac{1}{6} + \alpha\right)^7 \left(\frac{1}{2} - 3\alpha\right)^6 \left(\frac{1}{3} + 2\alpha\right)^{12}$$

Taking logs

$$\ln L(\alpha) = 7 \ln\left(\frac{1}{6} + \alpha\right) + 6 \ln\left(\frac{1}{2} - 3\alpha\right) + 12 \ln\left(\frac{1}{3} + 2\alpha\right)$$

[3]

ii) MLE:

$$\frac{d}{d\alpha} \ln L(\alpha) = \frac{7}{\left(\frac{1}{6} + \alpha\right)} - \frac{18}{\left(\frac{1}{2} - 3\alpha\right)} + \frac{24}{\left(\frac{1}{3} + 2\alpha\right)}$$

Equating this to zero, we have

$$\frac{7}{\left(\frac{1}{6} + \alpha\right)} - \frac{18}{\left(\frac{1}{2} - 3\alpha\right)} + \frac{24}{\left(\frac{1}{3} + 2\alpha\right)} = 0$$

$$\text{This gives } 7\left(\frac{1}{2} - 3\alpha\right)\left(\frac{1}{3} + 2\alpha\right) - 18\left(\frac{1}{6} + \alpha\right)\left(\frac{1}{3} + 2\alpha\right) + 24\left(\frac{1}{6} + \alpha\right)\left(\frac{1}{2} - 3\alpha\right) = 0$$

$$= (1+6\alpha) \left[\frac{7}{3}\left(\frac{1}{2} - 3\alpha\right) - \frac{18}{3}\left(\frac{1}{6} + \alpha\right) + \frac{24}{6}\left(\frac{1}{2} - 3\alpha\right) \right] = 0$$

$$\text{Thus, either } (1+6\alpha) = 0 \text{ or } \frac{7}{3}\left(\frac{1}{2} - 3\alpha\right) - \frac{18}{3}\left(\frac{1}{6} + \alpha\right) + \frac{24}{6}\left(\frac{1}{2} - 3\alpha\right) = 0 \text{ giving}$$

$$\alpha = -\frac{1}{6} \text{ or } \frac{13}{6} - 25\alpha = 0.$$

$$\text{That is } \alpha = -\frac{1}{6} = -0.0167 \text{ or } \alpha = \frac{13}{150} = 0.0867. [\text{roots of log likelihood equation}]$$

However, one of the roots $\alpha = -\frac{1}{6}$ is inadmissible as MLE since it does not lie in the range of α .

$$\frac{\partial^2 \ln L(\alpha)}{\partial \alpha^2} = -\frac{7}{\left(\frac{1}{6} + \alpha\right)^2} - \frac{54}{\left(\frac{1}{2} - 3\alpha\right)^2} - \frac{48}{\left(\frac{1}{3} + 2\alpha\right)^2} < 0 \text{ for } \alpha = 0.0867 \text{ confirming the maximum.}$$

[7]

iii) We have one unknown parameter, so we will use $E(X) = \bar{x}$.

$$E(X) = -1\left(\frac{1}{6} + \alpha\right) + 0\left(\frac{1}{2} - 3\alpha\right) + 1\left(\frac{1}{3} + 2\alpha\right) = \frac{1}{6} + \alpha$$

$$\text{From the data, we have: } \bar{x} = \frac{-1(7) + 0(6) + 1(12)}{25} = \frac{5}{25} = 0.2$$

$$\text{Therefore: } \frac{1}{6} + \alpha = 0.2 \Rightarrow \hat{\alpha} = \frac{1}{30} = 0.0333$$

[2]

[12 marks]

Solution 9:

i) Using the subscript 1 to refer to “control” and 2 for “with additive”, the test statistic is:

$$\frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{S_P \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1 + n_2 - 2}$$

Calculating the required values:

$$\bar{X}_1 = \frac{160.2}{10} = 16.02; \bar{X}_2 = \frac{176.8}{10} = 17.68$$

$$S_1^2 = \frac{1}{9} (2633.7 - 10 (16.02)^2) = 7.4773;$$

$$S_2^2 = \frac{1}{9} (3195.3 - 10 (17.68)^2) = 7.7196$$

$$\text{The pooled variance is: } S_P^2 = \frac{1}{18} (9 (7.4773) + 9 (7.7196)) = 7.5984$$

Hence, 95% confidence interval is:

$$(16.02 - 17.68) \pm 2.101 \sqrt{7.5984 \left(\frac{2}{10}\right)} = (-4.25, 0.93)$$

So the confidence interval is $(-4.25, 0.93)$.

Alternatively, the confidence interval for $\mu_2 - \mu_1$ is $(-0.93, 4.25)$.

Assumptions: The samples come from normal distributions with the same variance and that the samples are independent.

Comment: Since the confidence interval contains the value 0, we cannot say that the fuel additive significantly increases the mean mileage.

[6]

ii) The width of the confidence interval is:

$$2 \left(t_{2n-2; 2.5\%} \sqrt{\frac{((7.4773 + 7.7196)(n-1))}{2n-2} \cdot \frac{2}{n}} \right) = \frac{7.7966 t_{2n-2; 2.5\%}}{\sqrt{n}}$$

This must be less than 4, so using trial and error method we have:

$$\text{for } n = 16; \text{ width} = \frac{7.7966 (2.042)}{\sqrt{16}} = 3.98 < 4$$

The minimum sample size required is 16.

[4]

iii) No, we should not use a paired t-test. The cars were all similar but separate cars were used in the control and additive tests that are independent samples, which were not matched up in any way.

On the other hand, if independent samples are analyzed as though they were paired, then the results would be valid although they would be making inefficient use of the data due to the discarding of possible information about the means and variances of the two separate populations.

[2]

[12 marks]

Solution 10:

i) The hypotheses are:

H_0 : There is no association between yield of milk and horn type

H_1 : There is some association between yield of milk and horn type.

The expected frequencies are:

Horn \ Yield	Low	Medium	High
Curved	139.29	131.10	74.61
Straight	183.71	172.90	98.39

The test statistic is:

Horn \ Yield	Low	Medium	High
Curved	1.77	31.34	31.39
Straight	1.34	23.76	23.80

$$\sum \frac{(O-E)^2}{E} = 1.77 + 31.34 + 31.39 + 1.34 + 23.76 + 23.80 = 113.41$$

Degrees of freedom = (No of rows – 1) (No. of columns – 1) = (2 – 1) (3 – 1) = 2

Comparing the test statistic with the figures in the Tables for the χ^2_2 distribution, we see that this figure is much larger than 15.20 (0.5% level). Thus we have overwhelming evidence against the null hypothesis and conclude that there is some association between yield of milk and horn type.

[5]

ii) Using $\hat{p} = 0.1846$, we find the expected frequencies by multiplying the probabilities by 800. This gives the following table of frequencies expected by the genetic model:

Horn \ Yield	Low	Medium	High
Curved	147.68	73.84	108.72
Straight	172.29	246.13	51.33

For the chi-square goodness of fit test, the hypotheses are:

H_0 : The probabilities of each buffalo type conform to the genetic model

H_1 : The probabilities of each buffalo type do not conform to the genetic model.

The test statistic is:

Horn \ Yield	Low	Medium	High
Curved	0.36	0.63	1.88
Straight	0.11	0.34	0.03

$$\sum \frac{(O-E)^2}{E} = 0.36 + 0.63 + 1.88 + 0.11 + 0.34 + 0.03 = 3.35$$

Degrees of freedom = number of cells minus number of estimated parameters minus 1.

This gives us $6-1-1 = 4$ degrees of freedom.

Comparing the test statistic with the figures in the Tables for the χ^2_4 distribution, we see that it is close to 50% point. Hence, we do not have evidence to reject H_0 , and we conclude that the genetic model appears to fit the data well.

[5]

iii)

Test in part (i) indicated the existence of some association between milk yield and horn type but do not appear to fit the data well. Of the buffaloes with curved horns, there appears to be far too many with high yielding and far too few with medium yielding than we would expect under the assumption of independence.

Similarly, of the buffaloes with straight horns, there appears to be far too many with medium yielding and far too few with high yielding than we would expect under the assumption of independence.

However, the genetic model in part (ii) appears to compensate suitably for the above discrepancies, with the result that the actual number of buffaloes are closer to the numbers predicted by the model.

[2]

[12 marks]

Solution 11:

i)

$$S_{xx} = \sum_{i=1}^n x_i^2 - n\bar{x}^2 = 385 - 11 \left(\frac{55}{11} \right)^2 = 110$$

$$S_{yy} = \sum_{i=1}^n y_i^2 - ny^2 = 76,055 - 11 \left(\frac{701}{11} \right)^2 = 31,382$$

$$S_{xy} = \sum_{i=1}^n x_i y_i - n \bar{x} \bar{y} = 5,250 - 11(5)(63.7273) = 1,745$$

$$\text{We know that } \hat{\beta} = \frac{S_{xy}}{S_{xx}} = 1,745 / 110 = 15.8636$$

$$\text{And } \hat{\alpha} = \bar{y} - \hat{\beta} \bar{x} = \frac{701}{11} - 15.8636 \left(\frac{55}{11} \right) = -15.5909$$

Hence, the regression line is $\hat{y} = -15.5909 + 15.8636 x$

[2]

ii) Proportion of variability explained by the model is the coefficient of determination, R^2 , which is given by:

$$R^2 = \frac{S_{xy}^2}{S_{xx} S_{yy}} = \frac{1,745^2}{(110)(31,382)} = 88.21\%$$

This tells us that the model explains 88% of the variation in the NAVs. Since this leaves only 12% from other non-model sources, it would appear that the model is a very good fit to the data.

[2]

iii) Estimate of NAV at time 11 using the fitted regression line is given by

$$\hat{y} = -15.5909 + 15.8636(11) = 159$$

NAV at time 11 estimated by the model is lower than NAV at time 10 whereas we would expect the other way as per the trend of NAV growth seen for the past ten years. Hence, the model may not be suitable to predict future NAVs.

[2]

iv) The residuals, e_i , are calculated from the actual prices, y_i , and the predicted prices, $\hat{y}_i$:

$$e_i = y_i - \hat{y}_i$$

Using our regression line $\hat{y}_i = -15.5909 + 15.8636 x_i$ from part (i), we get:

$$\text{For } x = 3, \hat{y}_i = -15.5909 + 15.8636(3) = 32; e = 24 - 32 = -8$$

$$\text{For } x = 4, \hat{y}_i = -15.5909 + 15.8636(4) = 48; e = 32 - 48 = -16$$

$$\text{For } x = 5, \hat{y}_i = -15.5909 + 15.8636(5) = 64; e = 44 - 64 = -20$$

$$\text{For } x = 6, \hat{y}_i = -15.5909 + 15.8636(6) = 80; e = 58 - 80 = -22$$

$$\text{For } x = 7, \hat{y}_i = -15.5909 + 15.8636(7) = 95; e = 79 - 95 = -16$$

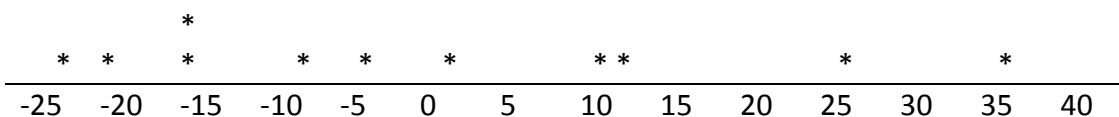
$$\text{For } x = 8, \hat{y}_i = -15.5909 + 15.8636(8) = 111; e = 106 - 111 = -5$$

[3]

v) The residuals in ascending order are:

(22) (20) (16) (16) (8) (5) 2 11 13 26 36

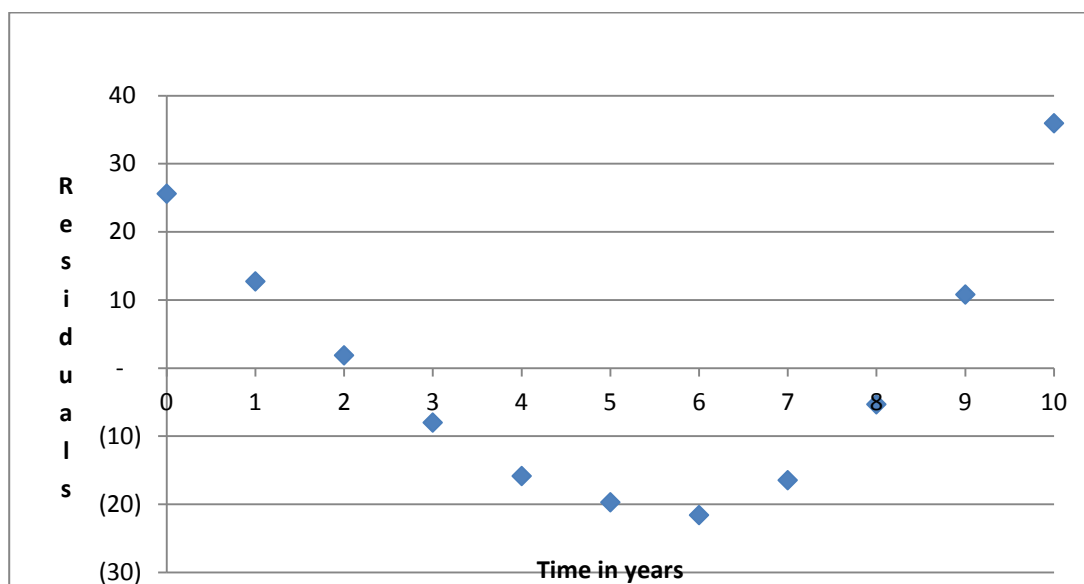
The dot plot is:



We would expect the dot plot to be normally distributed about zero and positive variance. This does not appear to be the case, but it is difficult to tell with such a small data set.

[2]

vi) The graph is:



Clearly, this is not pattern-less! The residuals are not independent of the time – this means that the linear model is definitely missing something and is not appropriate to the NAV data.

The NAV increases in an exponential (rather than linear) way. We should have used the log of the NAV against time instead.

[3]

[14 marks]

Solution 12:

i) H_0 : Each catalyst has the same efficiency.

H_1 : There is difference in the efficiency of at least one pair of catalysts.

Groups	Size	Sum	Sample mean	Square of sum
Catalyst (C1)	5	4.14	0.828	17.14
Catalyst (C2)	6	5.28	0.880	27.88
Catalyst (C3)	5	4.52	0.904	20.43

ANOVA

Source of Variation	Sum of Squares	df	Mean Sum of squares	F
Between Catalyts	0.0152	2	0.007588	4.403
Residual	0.0224	13	0.001723	
Total	0.0376	15		

At 5 % significance, $F_{2, 13}$ is 3.806. Hence, we conclude that there is significant difference in the efficiency of the catalysts.

[5]

ii) From the data it is seen that $\bar{y}_1. < \bar{y}_2. < \bar{y}_3.$

$$\hat{\sigma}^2 = SS_R / (n - k) = 0.001723$$

The table value $t_{0.05,13}$ is 1.771

$$\text{Hence, } t_{0.05,13} \times \hat{\sigma} \times \sqrt{\frac{1}{5} + \frac{1}{6}} = 1.771 \times \sqrt{0.001723} \times .606$$

$$= 1.771 \times 0.042 \times 0.606 = 0.045$$

$$\bar{y}_2. - \bar{y}_1. = 0.052.$$

This is greater than 0.045; hence using least significance approach, Catalyst 2 is better than the Catalyst 1.

We can compare $\bar{y}_3. - \bar{y}_2.$ because C1 can be ignored as C2 is better than C1

$$\bar{y}_3. - \bar{y}_2. = 0.024$$

and this is lesser than 0.045; hence using least significance approach, there is no difference between the efficiencies of Catalyst 2 and the Catalyst 3. There is no statistical advantage based on the given criteria to pay more for Catalyst 3.

Hence, C2 is recommended as the catalyst.

[3]

[8 marks]
